



INTERNSHIP REPORT

DATA SCIENCE MASTER VIRTUAL INTERNSHIP

Submitted

To

School of Computer Science and Engineering

IILM University, Greater Noida, U.P.

In partial fulfilment of the requirement of degree of

B.Tech CSE

By

MD RIHAN ALAM

Enrollment Number: 25SCS1003001381

Program: B.Tech CSE (AIML)

Semester: 3rd

Organization: EduSkills

Supported by: Siemens

Duration: June – August 2026

Batch: 2026–27

CANDIDATE’S DECLARATION

I, MD RIHAN ALAM, do hereby declare that the internship report titled “**DATA SCIENCE MASTER**

” has been completed by me to fulfil the requirement for the award of the degree of Bachelor of Technology in CSE. All the references have been quoted and I have not taken as such any material from any other source. I affirm to you that this report is my own and has not been submitted to any other institute for any degree/diploma requirement and further I shall be solely responsible for any kind of copyright violation in this regard.

Signature: _____

Student Name: MD RIHAN ALAM

Enrollment Number: 25SCS1003001381

ACKNOWLEDGEMENT

I am using this opportunity to express my gratitude to everyone who supported me throughout the Internship Program. I am thankful for their aspiring guidance, invaluable constructive criticism and friendly advice during this work. I am sincerely grateful to them for sharing their truthful and illuminating views on a number of issues related to this work.

I express my gratitude to my parents for their blessings.

Signature: _____

Student Name: MD RIHAN ALAM

Enrollment Number: 25SCS1003001381

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अखिल भारतीय तकनीकी शिक्षा परिषद्
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EduSkills®
Nation Building Through Skills



Certificate of Virtual Internship

This is to certify that

MD Rihan Alam

IILM University, Greater Noida

has successfully completed the 8-weeks

Data Science Master Virtual Internship

During June - August 2026

May this Internship learning propel you toward a bright and successful career.

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Certificate ID : 44d8bd5681ca0506fc21

Student ID: STU6a3ff95d5948a1782577501



GRADE- O (Outstanding): 90-100 | E (Excellent): 80-89 | A (Very Good): 70-79 | B (Good): 60-69 | C (Fair): 50-59 | D (Average): 40-49 | P (Pass): 30-39 | F (Fail): Below 30

*Internship Completion Certificate — reproduced from the
certificate provided by the student.*

4. PROJECT DESCRIPTION

4.1 INTRODUCTION

4.1.1 Executive Summary

The Data Science Master Virtual Internship was an eight-week beginner-level virtual internship completed through EduSkills during June–August 2026, with support from Siemens. The internship was intended to provide practical exposure alongside academic learning and to introduce the basic concepts, tools, and workflow used in Data Science. The learning covered the fundamental stages of working with data, from collection and cleaning to basic analysis, visualization, and the generation of simple insights.

During the internship, I developed a basic understanding of Data Science as a process of working with data to understand patterns, generate insights, and support better decisions. I was introduced to Python and a set of commonly used Python-based tools: Jupyter Notebook, NumPy, Pandas, Matplotlib, and Seaborn. The learning level remained introductory, with emphasis on understanding concepts and becoming familiar with the basic workflow rather than developing advanced Machine Learning or Deep Learning expertise.

The internship also helped me understand that useful analysis depends on properly prepared data. I learned the basic importance of checking datasets, identifying missing or inconsistent values, performing simple analysis, and using charts to make patterns easier to understand. Overall, the internship provided a foundation for further learning in Data Science, Artificial Intelligence, and Machine Learning.

4.1.2 Introduction to the Internship

The internship was undertaken as the Data Science Master Virtual Internship. The presentation supplied for the internship describes the program as an eight-week virtual internship designed around beginner-level Data Science concepts, tools, and workflows. It connects academic learning with practical exposure by introducing students to the process of handling and understanding data.

The internship was relevant to my B.Tech CSE (AIML) program because Data Science provides an important foundation for understanding data-driven computing

and future study in Artificial Intelligence and Machine Learning. At the end of the internship, the main learning outcome was not advanced specialization but a clearer understanding of the basic steps involved in a Data Science workflow and the tools used to carry them out.

Table 4.1: Internship Details

Item	Details
Student Name	MD Rihan Alam
Enrollment Number	25SCS1003001381
University	IILM University, Greater Noida
Program	B.Tech CSE (AIML)
Semester	3rd
Organization	EduSkills
Internship Title	Data Science Master Virtual Internship
Duration	8 Weeks
Period	June – August 2026
Supported By	Siemens

4.1.3 Importance of Data Science

Data Science is important because modern organizations work with large amounts of information and need ways to understand that information. The internship presentation summarizes Data Science through the relationship $\text{DATA} + \text{ANALYSIS} + \text{INSIGHTS} = \text{DATA SCIENCE}$. In practical terms, this means that useful work with data requires more than simply collecting values; the data must be checked, analyzed, visualized, and interpreted so that meaningful patterns can be understood.

At a beginner level, Data Science also helps students develop a structured way of thinking. Instead of looking at data as a collection of isolated values, a student learns to ask what the data represents, whether it is complete and consistent, what simple patterns are visible, and how those patterns can be communicated through graphs. These fundamentals are useful before moving to more advanced areas of Machine Learning and Artificial Intelligence.



Figure 4.1: Basic Data Science workflow represented in the internship presentation.

4.1.4 Purpose and Duration

The purpose of the internship was to build a basic foundation in Data Science through exposure to concepts and tools that can be used for data handling and analysis. The internship lasted eight weeks and was completed during June–August 2026. The program was virtual and was conducted through the EduSkills internship framework, with Siemens shown as the supporting organization in the provided certificate and presentation.

4.2 ORGANIZATION PROFILE

EduSkills is presented in the internship reference material as a provider of industry-oriented learning opportunities and virtual internships. The internship format is intended to help students gain practical exposure alongside academic learning. In the context of this internship, the Data Science program introduced beginner-level concepts, tools, and workflows rather than advanced professional specialization.

The virtual internship model was useful for connecting classroom concepts with a structured learning experience. It provided an opportunity to become familiar with the language and tools commonly associated with introductory Data Science, including Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, and Seaborn. The program was supported by Siemens, as shown in both the presentation and the internship completion certificate.

From a student-learning perspective, the organization's role in this internship can therefore be understood as providing an industry-oriented environment in which foundational Data Science knowledge could be explored. The main benefit was exposure to a practical workflow and tools that can be developed further through future study and projects.

4.2.1 Role of the Internship in Skill Development

- Introduced the basic meaning and application of Data Science.
- Provided exposure to basic data collection, cleaning, handling, and analysis.
- Introduced Python-based tools used for working with data.
- Developed awareness of basic visualization methods and their purposes.

- Created a foundation for future learning in AI, Machine Learning, and Data Science.

4.3 PROBLEM STATEMENT

The central learning problem addressed by the internship was how a beginner can move from raw or collected data toward understandable information and simple insights. Data by itself may contain rows, columns, different types of values, missing entries, or inconsistencies. Without basic checking and analysis, it can be difficult to understand what the data is showing.

The internship therefore focused on understanding a simple and repeatable approach: collect or load data, inspect it, identify missing or inconsistent values, clean the data, perform basic analysis, create suitable visualizations, and use the resulting patterns to form simple insights. This problem statement is intentionally limited to the beginner-level scope shown in the internship presentation and does not assume an advanced predictive model or a specific real-world dataset.

A related learning problem was becoming familiar with the tools required to perform these steps. For a beginner, Python programming, notebook-based experimentation, numerical arrays, tabular data, and visualization libraries can initially appear as separate topics. The internship helped connect them as parts of a single Data Science workflow.

4.3.1 Data Science Fundamentals Covered

- Introduction to Data Science and its role in working with data.
- Types of data at a beginner level.
- Data collection and basic data handling.
- Data cleaning and awareness of missing values.
- Basic Data Analysis using simple ways to explore datasets.
- Exploratory Data Analysis (EDA) at a beginner level.
- Data visualization using common chart types.
- Introduction to Machine Learning at a beginner level.

4.3.2 Understanding Data Collection and Cleaning

Data collection is the stage in which information required for analysis is obtained and organized. The internship introduced the idea that data must be understood before

it can be analyzed. This includes recognizing what the rows and columns represent and checking whether the information appears complete and consistent.

Data cleaning is the basic process of identifying problems in data and preparing it for analysis. The presentation specifically mentions missing values and inconsistent values. At the beginner level, the important learning was to recognize that the quality of the input data affects the usefulness of later analysis. The internship did not claim advanced data-engineering techniques; the focus remained on understanding the purpose of cleaning and the need to inspect data before analysis.

4.4 PROJECT OBJECTIVES

The objectives of the internship, based on the provided presentation, were:

- Understand what Data Science is and where it is used.
- Understand basic data collection, cleaning, and handling.
- Learn simple ways to explore and understand datasets.
- Create and interpret basic charts and graphs.
- Become familiar with beginner-level Data Science libraries.
- Develop a base for further learning in Artificial Intelligence, Machine Learning, and Data Science.

4.4.1 Learning-Oriented Objectives

In addition to the direct objectives stated in the presentation, the internship was useful in developing a structured learning approach. I aimed to understand not only individual tools but also how the tools fit into the overall workflow. The objective was therefore to become comfortable enough with fundamental concepts to continue learning independently rather than to claim mastery of advanced techniques.

4.5 SCOPE OF THE PROJECT

The scope of this internship report is limited to the beginner-level Data Science learning described in the supplied presentation and certificate. It covers the basic workflow of working with data, the introductory concepts learned, and the Python-based tools explored during the eight-week virtual internship.

Table 4.5: Scope and reporting boundaries

Included in Scope	Not Claimed in This Report
Introduction to Data Science	Advanced Machine Learning expertise

Data collection and basic handling	Advanced Deep Learning
Data cleaning and missing values	Advanced statistical modelling
Basic Data Analysis and EDA	Fabricated datasets or performance results
Data visualization	Unreported projects or achievements
Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, Seaborn	Technologies not present in the reference material
Introduction to Machine Learning	Specific model accuracy, scores, or benchmarks not provided

This boundary is important because the internship was introductory. The report therefore emphasizes understanding, familiarity, and foundation-building. Any future development of these skills would require additional coursework, practice, datasets, projects, and deeper study.

4.6 TECHNOLOGIES AND TOOLS USED

The internship presentation identifies six principal tools and technologies. Their roles are described below at the same basic level used in the reference material.

Table 4.2: Tools and Their Basic Purpose

Tool / Technology	Basic Purpose
Python	Programming language used for data analysis.
Jupyter Notebook	Interactive environment for writing and testing code.
NumPy	Basic numerical computing and arrays.
Pandas	Working with tables and structured datasets.
Matplotlib	Creating basic charts and graphs.
Seaborn	Making statistical visualizations more easily.

4.6.1 Python

Python was introduced as the programming language used for data analysis. In this internship, the focus was on becoming familiar with Python as part of the Data Science workflow. Python provides the programming environment in which data can be loaded, inspected, processed, analyzed, and visualized with supporting libraries.

4.6.2 Jupyter Notebook

Jupyter Notebook was presented as an interactive environment for writing and testing code. For a beginner, a notebook is useful because code and its output can be

viewed together, making it easier to experiment with small steps and observe the effect of each operation while learning.

4.6.3 NumPy

NumPy was introduced for basic numerical computing and arrays. At the learning level of this internship, the important concept was that numerical data can be represented and manipulated efficiently using array-oriented operations.

4.6.4 Pandas

Pandas was introduced for working with tables and structured datasets. It is especially relevant to the basic Data Science workflow because datasets are often organized into rows and columns. The internship focused on fundamental familiarity with handling such tabular data.

4.6.5 Matplotlib

Matplotlib was introduced as a tool for creating basic charts and graphs. Visualization is an important part of the workflow because a graph can make patterns, comparisons, and changes easier to communicate than a table alone.

4.6.6 Seaborn

Seaborn was introduced for making statistical visualizations more easily. The internship included it as part of the beginner-level Python visualization toolkit, without claiming advanced statistical visualization expertise.

4.7 SYSTEM ARCHITECTURE

For this internship, the term “system architecture” is used to represent the conceptual flow of a basic Data Science task rather than a software product architecture. The internship presentation shows Data Science as a sequence beginning with data collection and moving through cleaning, analysis, visualization, and insights.

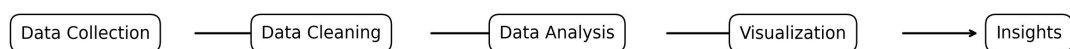


Figure 4.4: Conceptual system architecture / workflow for the internship's Data Science learning.

The workflow is sequential at a conceptual level. Data is first collected or loaded. It is then checked and cleaned so that obvious missing or inconsistent values can be identified. After cleaning, basic analysis is performed. Visualization is then used to communicate patterns and trends, leading to simple insights. This structure provides a beginner-friendly mental model for organizing Data Science work.

4.7.1 Components of the Architecture

1. Data Collection: obtain or load the information to be studied.
2. Data Cleaning: identify missing or inconsistent values and prepare the data.
3. Data Analysis: use simple methods to explore and understand the dataset.
4. Visualization: represent information using suitable charts and graphs.
5. Insights: identify understandable patterns or observations from the analyzed data.

4.8 METHODOLOGY

The methodology followed in the internship can be understood as a basic Data Science learning and analysis process. The presentation gives the sequence “Load Data → Check Data → Clean Data → Analyze Data → Create Graphs → Find Insights.” The following subsections explain each stage without extending beyond the level supported by the reference material.

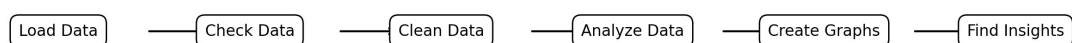


Figure 4.2: Basic Data Analysis Process shown in the internship presentation.

4.8.1 Load and Check Data

The first step is to load or access the dataset and understand its structure. This involves reading the rows and columns and identifying what kind of information is represented. Checking data before analysis is important because it provides an initial understanding of what can be studied and whether the dataset contains obvious problems.

- Read the rows and columns of a dataset.

- Observe the structure and types of information present.
- Identify missing or inconsistent values.

4.8.2 Clean Data

After checking the data, the next step is cleaning. At the beginner level covered in the internship, this means being aware of missing or inconsistent values and preparing the data for basic analysis. Cleaning is treated as a necessary part of the workflow rather than an optional step, because analysis on poorly understood data may lead to unclear or misleading observations.

4.8.3 Analyze Data

Basic Data Analysis involves simple ways of exploring and understanding the dataset. The presentation specifically mentions using simple statistics to understand data. The objective is to learn what the data contains and to identify straightforward patterns or differences rather than to perform advanced statistical modelling.

4.8.4 Exploratory Data Analysis (EDA)

Exploratory Data Analysis was covered at a beginner level. In this context, EDA means examining a dataset to understand its structure, values, and basic patterns before drawing conclusions. It connects data checking, simple analysis, and visualization. The internship introduced the concept rather than claiming advanced EDA techniques.

4.8.5 Data Visualization

Visualization was an important part of the learning because it makes data easier to understand and helps identify patterns and trends. The presentation identifies several basic chart types and their common purposes.

Table 4.3: Basic Visualization Types

Visualization	Purpose
Bar Chart	Compare categories
Line Chart	Show change over time
Pie Chart	Show proportions
Histogram	Understand distribution
Scatter Plot	Observe relationships

Bar Chart	Compare categories
Line Chart	Show change over time
Pie Chart	Show proportions
Histogram	Understand distribution
Scatter Plot	Observe relationships

Figure 4.3: Basic visualization types and their purposes, based on the internship presentation.

4.8.6 Introduction to Machine Learning

The internship included an introduction to Machine Learning as a beginner-level concept. At this stage, the learning outcome was awareness of Machine Learning as an area connected to Data Science and future AI learning. The internship did not establish advanced Machine Learning expertise, model development experience, performance benchmarks, or specialized algorithms, and therefore none of those are claimed in this report.

4.8.7 Basic Data Science Workflow in Practice

The overall methodology can be summarized as a cycle of understanding data and progressively making it more useful for interpretation. First, the data is obtained and inspected. Second, basic cleaning is carried out. Third, simple analysis is performed. Fourth, the results are represented visually. Finally, the patterns observed through the analysis and visualization can be expressed as simple insights. This workflow was one of the key conceptual outcomes of the internship.

4.9 EXPECTED OUTCOMES

The expected outcomes of the internship are learning outcomes rather than claims of advanced technical specialization. Based on the presentation, the internship helped

develop familiarity with Data Science concepts, Python-based tools, data handling, analysis, and visualization. It also provided a foundation for future study.

4.9.1 Learning Experience

The internship helped me build a basic understanding of the Data Science field and become familiar with the way data is handled in an introductory workflow. I learned that Data Science involves more than coding; it requires understanding data, checking its quality, analyzing it carefully, and communicating observations through suitable visualizations.

I also became familiar with Python-based Data Science tools. Working conceptually with Jupyter Notebook, NumPy, Pandas, Matplotlib, and Seaborn helped me understand the relationship between programming, data handling, numerical operations, and visualization. This experience improved my confidence in approaching data-related tasks at a beginner level.

The internship further developed my analytical thinking by encouraging me to look at data systematically. The learning experience has given me a foundation from which I can continue studying Artificial Intelligence, Machine Learning, and Data Science in greater depth.

4.9.2 Technical Skills Gained

- Basic Python for Data Science.
- Basic data handling.
- Data cleaning and awareness of missing values.
- Basic Data Analysis.
- Data visualization.
- Pandas and NumPy fundamentals.
- Basic familiarity with Jupyter Notebook, Matplotlib, and Seaborn.

4.9.3 Professional Skills Gained

- Problem Solving.
- Analytical Thinking.
- Self-Learning.
- Time Management.
- Consistency.

- Technical Communication.

Table 4.4: Technical and Professional Skills Gained

Category	Skills
Technical	Basic Python; Basic Data Handling; Data Cleaning; Basic Data Analysis; Data Visualization; Pandas & NumPy fundamentals
Professional	Problem Solving; Analytical Thinking; Self-Learning; Time Management; Consistency; Technical Communication

4.9.4 Challenges and Learning

As a beginner, understanding datasets and their structure was an important learning challenge. Data can contain different types of values, missing entries, or inconsistencies, and it is necessary to inspect the information before trying to analyze it. This reinforced the importance of data checking and cleaning.

Another challenge was becoming familiar with several new tools at the same time. Python, Jupyter Notebook, NumPy, Pandas, Matplotlib, and Seaborn each have different roles, and understanding how they fit together required consistent practice and self-learning. The internship helped reduce this difficulty by presenting them as parts of a connected workflow.

Interpreting visualizations was also an important learning point. A chart is useful only when its purpose and pattern are understood correctly. Learning the basic purposes of bar charts, line charts, pie charts, histograms, and scatter plots helped me understand that different visualizations communicate different types of information.

These challenges were constructive because they encouraged a systematic learning approach. Instead of trying to learn advanced techniques immediately, I focused on understanding the fundamentals first and building a base for future learning.

4.9.5 Overall Learning Outcome

The overall outcome of the internship was a beginner-level foundation in Data Science. I developed awareness of the basic workflow, became familiar with Python-based tools, and learned how data can be checked, cleaned, analyzed, and visualized.

The experience also improved my confidence and analytical thinking while working with data.

The internship has encouraged me to continue exploring AI, Machine Learning, and Data Science. Further development will require deeper study and more practical work, but the fundamental concepts learned during this internship provide a useful starting point.

4.10 CERTIFICATES OF COMPLETION AND OTHER COMMUNICATION MAIL PROOF

The internship completion certificate provided with the submission materials is attached in this report in its original informational form. The certificate identifies the internship as the “Data Science Master Virtual Internship,” states that it was an eight-week virtual internship during June–August 2026, identifies EduSkills and IILM University, and shows Siemens as the supporting organization.

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Figure 4.4: Internship completion certificate provided by the student

4.10.1 Communication Mail Proof



EduSkills Fou... Jul 28

to me ▾



Dear MD Rihan Alam,



Congratulations!

You have successfully completed **10 weeks Siemens Data Science Master** internship with EduSkills Foundation. Your certificate is ready – please log in to your profile and download it.



Steps to Download Your Certificate:

1. Login at: <https://aicte-internship.eduskillsfoundation.org/>
2. Go to the **Certificate Section** from your sidebar
3. Click **Download Certificate**

Please post your certificate on your

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